

# Jie Shen

647-510-8128 | [jie22shen@gmail.com](mailto:jie22shen@gmail.com) | [website](#)

 [Jie Shen](#) |  [shenjie9](#) |  [jie22shen](#)

Mississauga, Ontario - L4Z4E1, Canada

## OBJECTIVE

Highly motivated, detail-oriented 4th-year university student pursuing a Bachelor of Science (Honours) in Mathematics, Statistics, and Computer Science. Seeking an internship in data science or computer science with a focus on machine learning to leverage advanced analytical skills, strong problem-solving abilities, and a solid foundation in quantitative methods.

## EXPERIENCE

- **City of Mississauga** July 2018 - March 2019  
*Summer Camp Leader* Mississauga, Canada
  - Developed team management and communication skills, key for collaborative project work in tech and data environments.
  - Designed structured activities, reflecting abilities in planning, critical thinking, and goal-oriented execution.

## EDUCATION

- **University of Toronto, Mississauga** September 2022 - Expected June 2027  
*Honours in Mathematics and Statistics, Minor in Computer Science* Mississauga, Canada

## PROJECTS

- **Rider-Driver UBER Interface**  
*Tools: [Python, data analysis packages (Pandas, NumPy, SciPy), visualization tools (Matplotlib)]*
  - Developed a ride-sharing simulation using Python to model ride allocation in a grid-based city.
  - Implemented an event-driven architecture leveraging priority queues to manage event scheduling and execution.
  - Employed data collection and analysis framework within the simulation, capturing detailed metrics like rider wait times, driver idle periods, and trip distances, facilitating statistical analysis and performance optimization.
  - Applied object-oriented design principles, ensuring modularity, scalability, and re-usability of core components.
- **Mobile Call Data Management and Visualization System**  
*Tools: [Python, Pygame, JSON, Pytest, Datetime Module, Object-oriented Design]*
  - Designed and implemented a mobile call tracking system using object-oriented modules and principles of inheritance and composition to manage customer data, call events, and billing information.
  - Applied data parsing techniques to process large datasets in JSON format and implemented algorithms to handle various contract types, ensuring accurate customer billing and contract management.
  - Integrated a visualization tool using Pygame to map call activity onto a geographic representation of Toronto, allowing users to explore customer calling patterns and network coverage interactively.
- **Interactive Treemap Visualization Tool for Hierarchical Data Representation**  
*Tools: [Python, Pygame, OS Module, CSV Parsing, Recursive Algorithms, Exploratory Data Analysis (EDA)]*
  - Engineered an interactive treemap visualization tool to model and analyze hierarchical data structures, such as file systems and categorized research datasets.
  - Employed Pygame to render dynamic, real-time visualizations, proportionally scaling data nodes based on their size, and offering users an intuitive, interactive interface.
  - Extended the system to support multiple datasets, including both file system visualizations and research paper categorizations, demonstrating versatility in handling complex hierarchical data models.
- **Othello Game Engine with AI-Based Decision Models**  
*Tools: [Java, OOP, Strategy Pattern, Git, Algorithm Optimization, JUnit]*
  - Engineered a modular Othello game engine supporting both human and AI players, with a focus on maintainable and extensible design using object-oriented principles and Strategy design pattern.
  - Developed automated decision-making algorithms for AI players, implementing Greedy and Random strategies to simulate competitive gameplay and enable various match configurations.
  - Integrated rigorous JUnit testing protocols, achieving thorough test coverage and ensuring alignment with project specifications and edge-case handling, resulting in a robust codebase.

## SKILLS

- **Programming Languages:** Python, Java, HTML5, CSS, JavaScript
- **Data Analysis/Visualization Tools:** R (programming language), SQL, Power BI, NumPy, Matplotlib, Microsoft Excel
- **Mathematical Techniques and Technical Skills:** Linear Algebra, Calculus, Differential Equations, Discrete Mathematics, Number Theory, Descriptive Statistics, Graphic Design, Robotics,  $\LaTeX$